Principle of Communication System	
Lab #:	**08**
Lab Title:	Sampling Theorem
Submitted by:	

Names	**Registration #**
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Rubrics name & number		**Marks**	
		In-Lab	**Post-Lab**
Engineering Knowledge	***R2: Use of Engineering Knowledge and follow Experiment Procedures:*** *Ability to follow experimental procedures, control variables, and record procedural steps on lab report.*		
Problem Analysis	***R6: Experimental Data Analysis:*** *Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.*		
Design	***R8: Best Coding Standards:*** *Ability to follow the coding standards and programming practices.*		
Modern Tools Usage	***R9: Understand Tools:*** *Ability to describe and explain the principles behind and applicability of engineering tools.*		
Individual and Teamwork	***R12: Individual Work Contributions:*** *Ability to carry out individual responsibilities.*		
	R13: Management of Team Work: *Ability to appreciate, understand and work with multidisciplinary team members.*		

Rubrics #	**R2**	**R6**	**R8**	**R9**	**R12**	**R13**
In Lab						
Post- Lab						

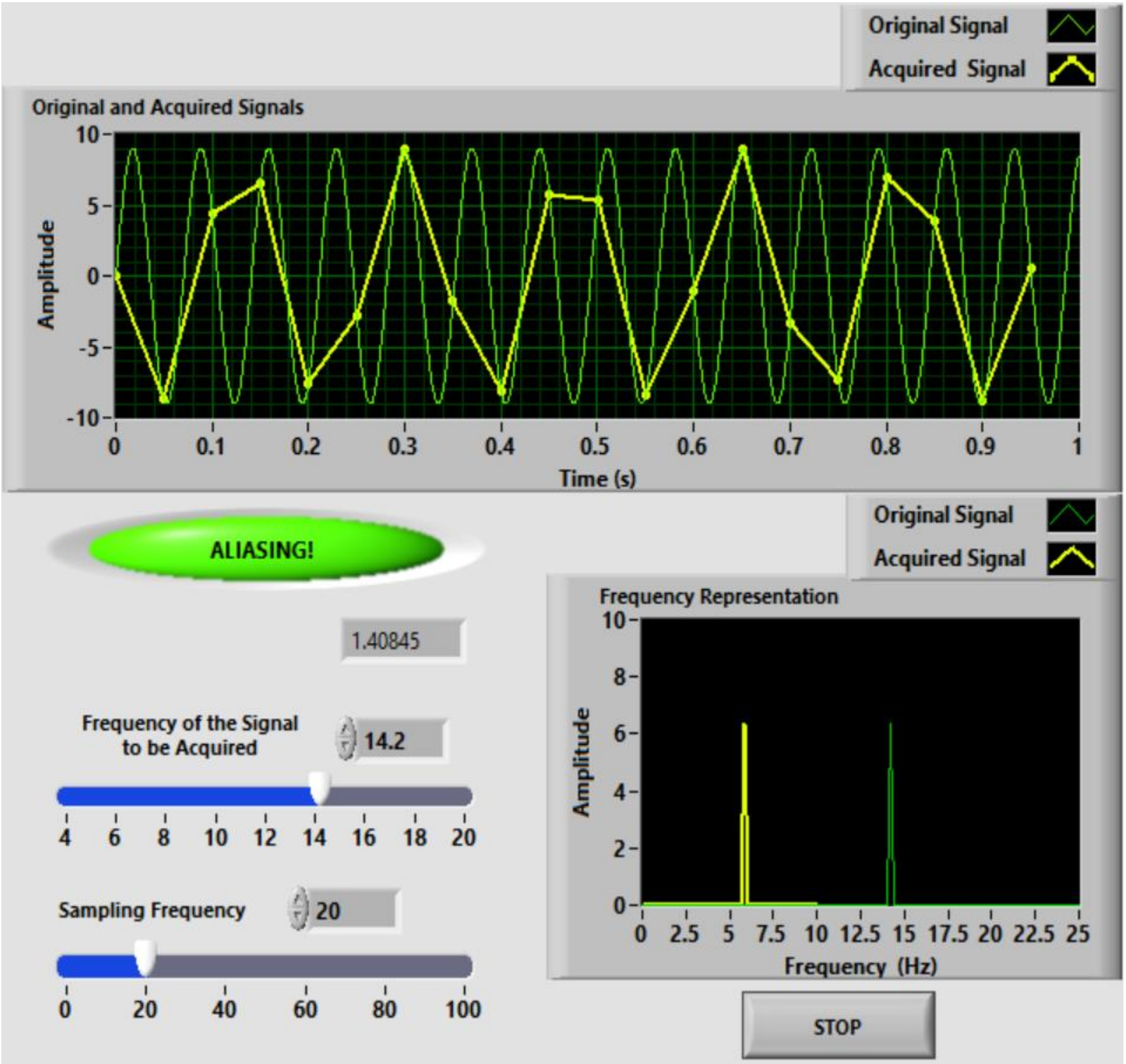
Lab 8: Sampling Theorem

Objective:

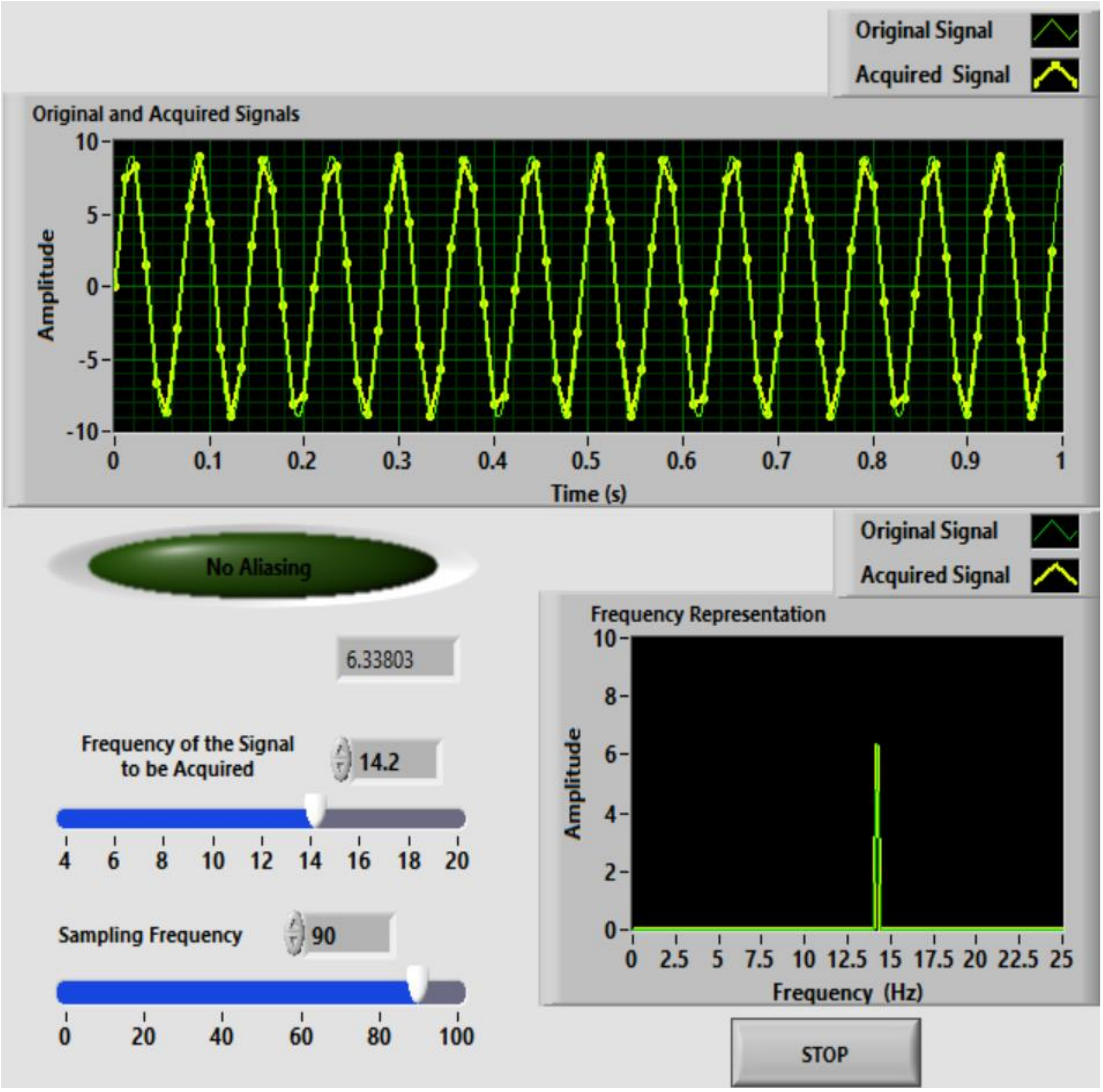
Observe the sampling theorem concept by varying the simulated signal frequency and sampling frequency and to observe the aliasing effect by varying the simulated frequency.

FRONT PANEL:

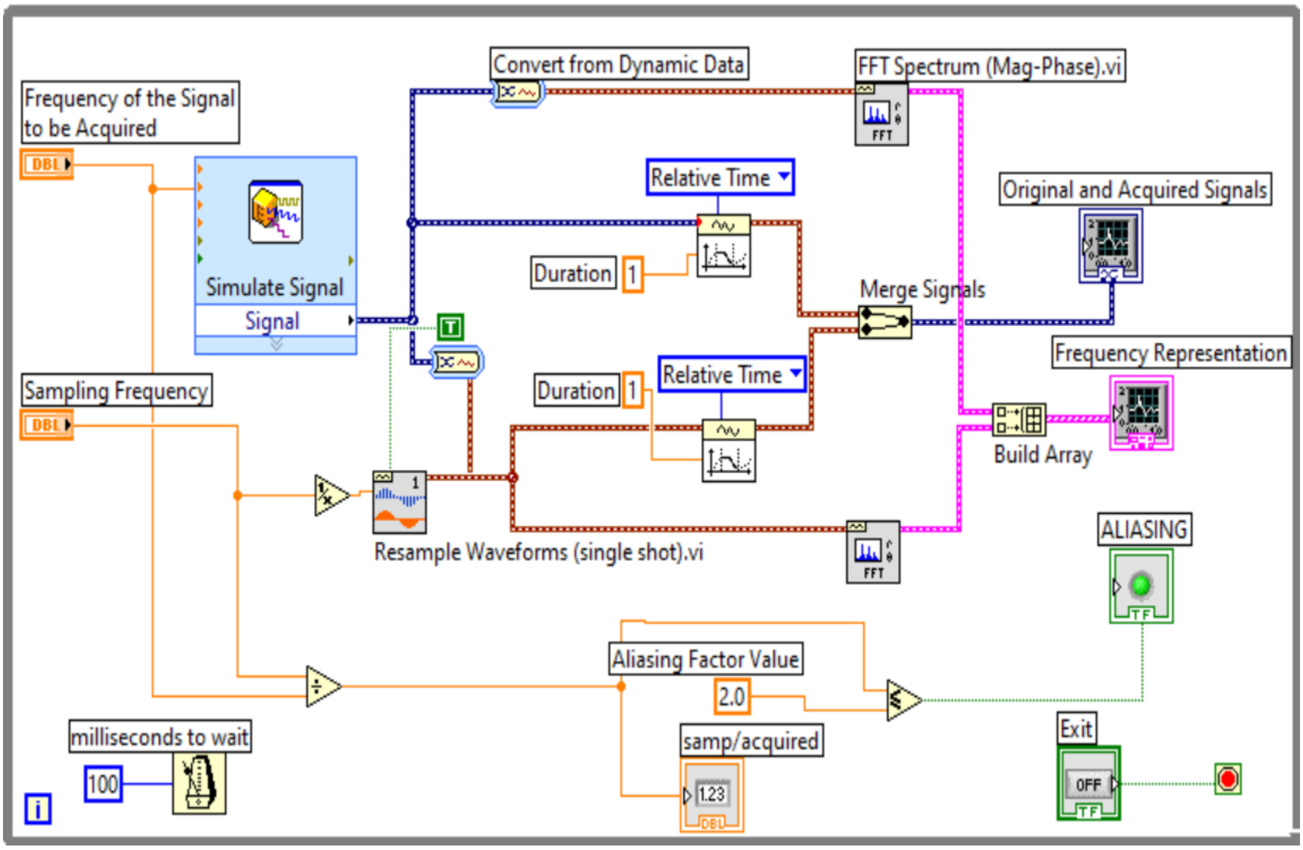
<u>With Alaising:</u>



Without Alaising:



Block Diagram:



Conclusion:

This demonstrated the principles of the Sampling Theorem, highlighting the critical relationship between sampling frequency and signal reconstruction. By observing the effects of varying sampling rates, we confirmed that a sampling frequency of at least twice the maximum signal frequency is essential to avoid aliasing and ensure accurate signal representation. This reinforces the importance of selecting appropriate sampling frequencies in practical applications, such as telecommunications and audio processing. Overall, the lab provided valuable insights into the implications of the Sampling Theorem in real-world scenarios.